out specialized types of metabolism (Figure 15.4). Plants are thus a mixture of older, mature cells and young, dividing cells.

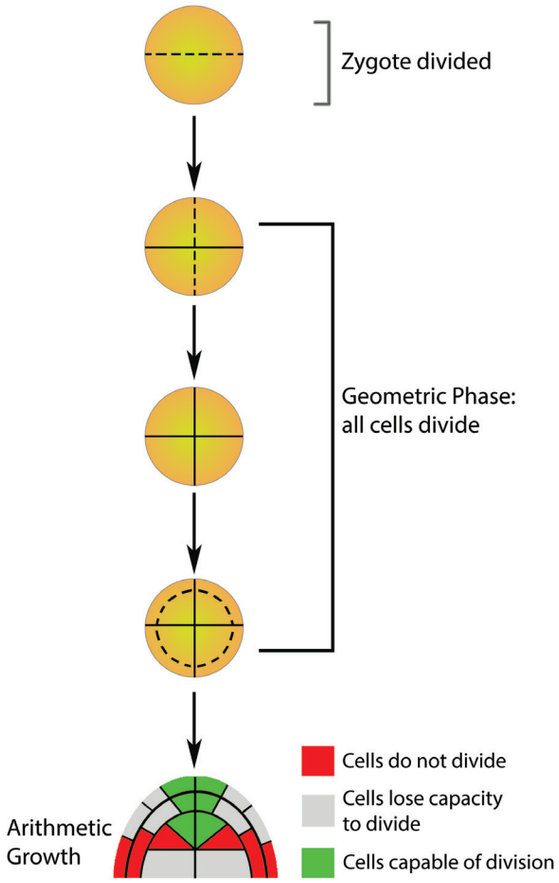


Figure 15.4: Arithmetic and geometric growth of embryo

In figure 15.5, two leaves A and B are drawn at a particular time. Then $A^1$and B^1 are drawn after a given time. A and B = Area of leaves at a particular time. A^1 and B^1 = Area of leaves after a given time. (A^1-A) and (B^1-B) represents an absolute increase in area in the given time. Leaf A increases from 5 cm^2 to 10 cm^2; 5 cm^2 in a given time. Leaf B increases from 50 cm^2 to 55 cm^2 ; 5 cm^2 in a given time. Hence, both leaves A and B increase their area by 5 cm^2 in a given time. This is absolute growth. Relative growth is faster in leaf A because of initial small size. It decreases with time.

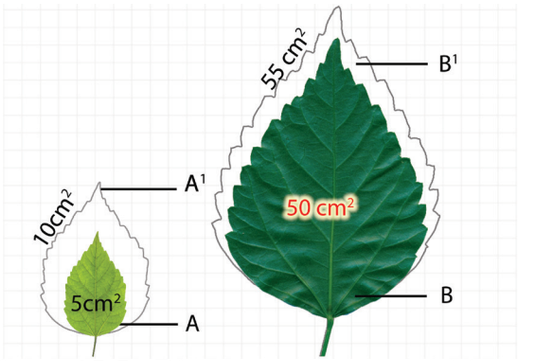


Figure 15.5: Diagrammatic comparision of absolute and relative growth rates

Measurement of Growth

Experiment: 1. Arc auxanometer:

The increase in the length of the stem tip can easily be measured by an arc auxanometer which consists of a small pulley to the axis of which is attached a long pointer sliding over a graduated arc. A thread one end of which is tied to the stem tip and another end to a weight passes over the pulley tightly. As soon as the stem tip increases in length, the pulley moves and the pointer slide over the graduated arc (Figure 15.6). The reading is taken. The actual increase in the length of the stem is then calculated by knowing the length of the pointer and the radius of the pulley. If the distance travelled by the pointer is 10 and the radius of the pulley is 4 inches and the length of the pint is 20 inches, the actual grown is measured as follows:

$$\text{Actual growth in length} = \frac{\text{Distance travelled by the pointer} \times \text{radius of the pulley}}{\text{Length of the pointer.}}$$

For example,

Actual growth in length =

$(10 \times 4 \text{ inches})/ 20 \text{ inches} = 2 \text{ inches}$

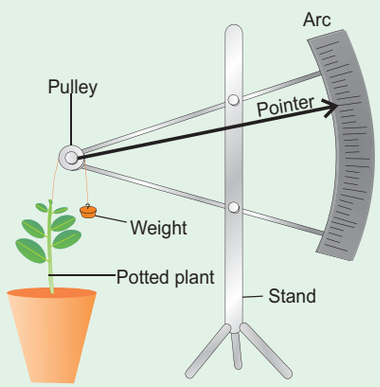


Figure 15.6: Arc auxanometer

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